

CareFlow Health Management System

CSI-4999 Senior Capstone - Jawad
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Video link

Project Description:

This project aims to create a healthcare management system intended to simplify operations across multiple medical locations. The system centralizes the management of clinics, staff, patients, billing, and medical records into a single secure platform. By replacing fragmented and paper-based workflows, the application aims to improve organizational efficiency, data accuracy, and accessibility for healthcare providers. The system is designed to support role-based access, scalable usage across multiple locations, and secure handling of sensitive patient information, making it suitable for small clinics, private practices, and healthcare organizations.

Target Audience:

The primary target audience for this healthcare management system includes small medical clinics, private medical practices, and larger healthcare organizations. These users would benefit from a centralized platform to manage multiple clinic locations, personnel, patient records, billing, and documentation. The system is especially suited for organizations seeking to replace manual hard copy based paperwork with a digital system.

Revenue Streams:

The healthcare management system will generate revenue through a subscription-based model offered on a monthly or yearly basis. Pricing will be tiered based on the number of staff members and patient users supported by the system, allowing organizations of different sizes to select a plan that fits their needs. Additional revenue may be generated through optional add-on features, such as expanded storage, advanced reporting, or enhanced billing tools, providing flexibility and scalability for growing healthcare organizations.

Similar Existing System:

CPR is a platform designed to replace paper charts by capturing, storing, and retrieving patient health information across care networks. Our system will allow users to manage multiple locations as well as information.

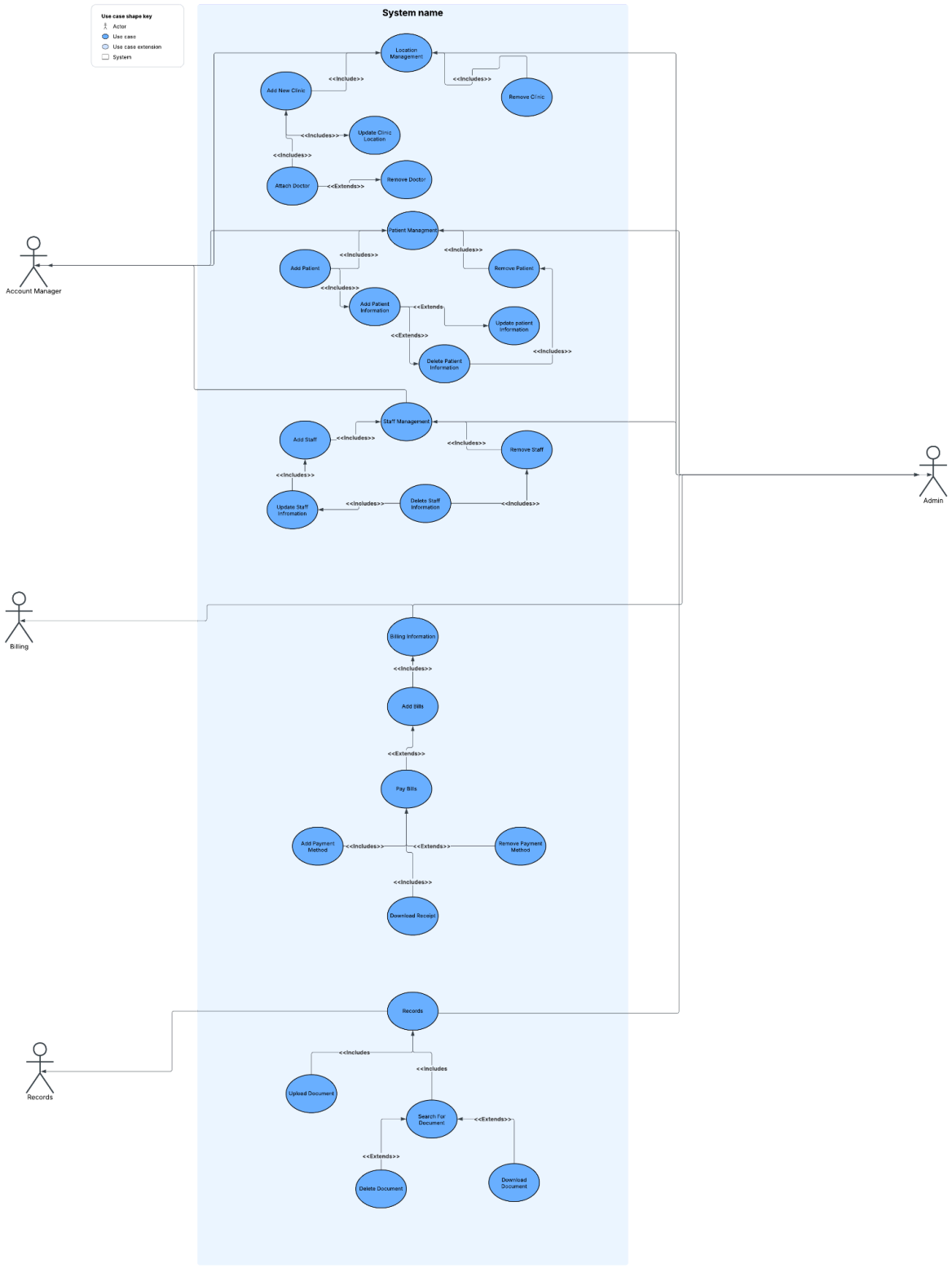
In contrast, our healthcare management system is designed as a broader operational platform. In addition to managing patient records, it supports multi-location clinic management, staff assignment, billing, payments, and document handling within a single integrated system. This approach allows healthcare organizations to manage both administrative and clinical workflows together

Subsystems and Features:

The healthcare management system is divided into five core subsystems. Each subsystem focuses on a specific operational area and includes multiple features that interact directly with a database through distinct use cases. These subsystems include *Location Management, Patient Management, Staff Management, Billing Information*, and *Records Management*. Together, they provide support for managing clinic locations, personnel, patient data, billing processes, and medical documentation, resulting in a system with over twenty distinct use cases.

- **Location Management**
    - Add new Clinic Locations
    - Remove (old) Clinic Locations
    - Update Clinic Locations
    - Attach Doctor to location
    - Remove Doctor from location
- **Patient Management**
    - Add patient
    - Remove patient
    - Add patient information
    - Update patient information
    - Delete patient information
- **Staff Management**
    - Add staff
    - Remove staff
    - Update staff information
    - Delete staff information
- **Billing Information**
    - Add bills
    - Pay bill
    - Add payment method
    - Remove payment method
    - Download Receipt
- **Records**
    - Upload document
    - Search for document
    - Download document
    - Delete document

Use Case Diagram:



20 Features of the System:

- Update patient records
- Add patient records
- Remove patient records
- Add patients
- Update patient information
- Remove patient
- Add staff
- Update staff information
- Remove staff
- Add clinic
- Attach head doctor to clinic
- Update clinic info
- Remove clinic
- View bills
- Pay bills
- Add payment method
- Remove payment method
- Download receipt
- Upload document
- Search document
- View document
- Download document
- Delete document

Users on the System:

The system is designed to support multiple user roles. The primary user types include *Administrators, Billing users, Records users*, and *Patient users*. Administrators have full access to all system features, including managing locations, staff, and system configuration. Billing users are limited to billing-related functions such as viewing and processing payments. Records users can manage medical documents and patient records, while Patient users are responsible for adding, updating, and maintaining patient information. Each role is granted access only to the features necessary for their responsibilities.

Tables and ERD (subject to change):

1. **ClinicLocation**
    - location_id (PK)
    - name, address, city, state, zip, phone
    - status (active/inactive)
    - Relationship ideas: One location has many staff assignments, many patients.

2. **Staff**
    - staff_id (PK)
    - first_name, last_name, email, phone

- role (admin/billing/records/patients)
- Active_flag

3. **StaffLocationAssignment** (handles “attach/remove doctor/staff to location”)
    - assignment_id (PK)
    - staff_id (FK → Staff)
    - location_id (FK → ClinicLocation)
    - assignment_role (e.g., “Head Doctor”, “Nurse”, “Reception”)
    - start_date, end_date (nullable)
    - Relationship ideas: Many-to-many between Staff and ClinicLocation.

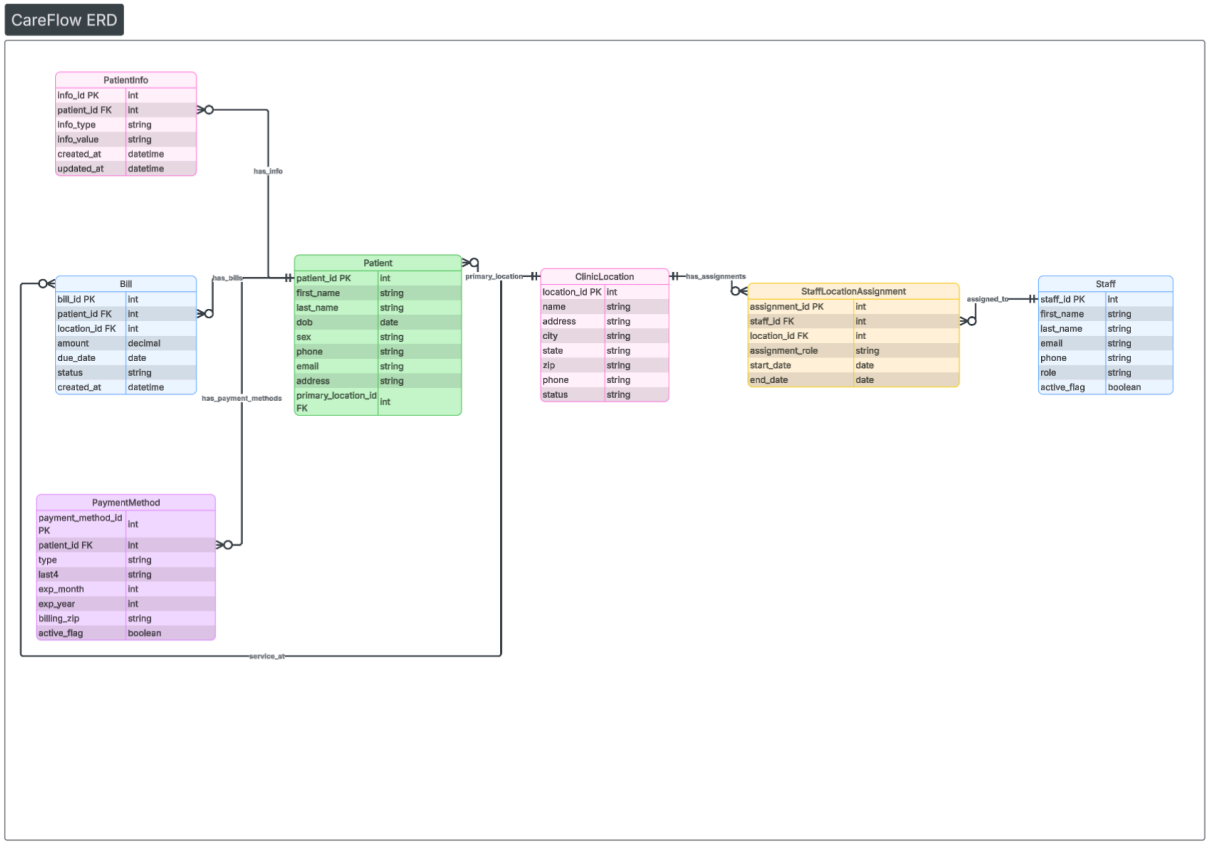
4. **Patient** (main table)
    - patient_id (PK)
    - first_name, last_name, dob, sex
    - phone, email, address
    - primary_location_id (FK → ClinicLocation)

5. **PatientInfo** (supports “add/update/delete patient information”)
    - info_id (PK)
    - patient_id (FK → Patient)
    - info_type (allergy, condition, medication, note, emergency_contact, etc.)
    - info_value (text)
    - created_at, updated_at
    - Relationship: One Patient has many PatientInfo entries.

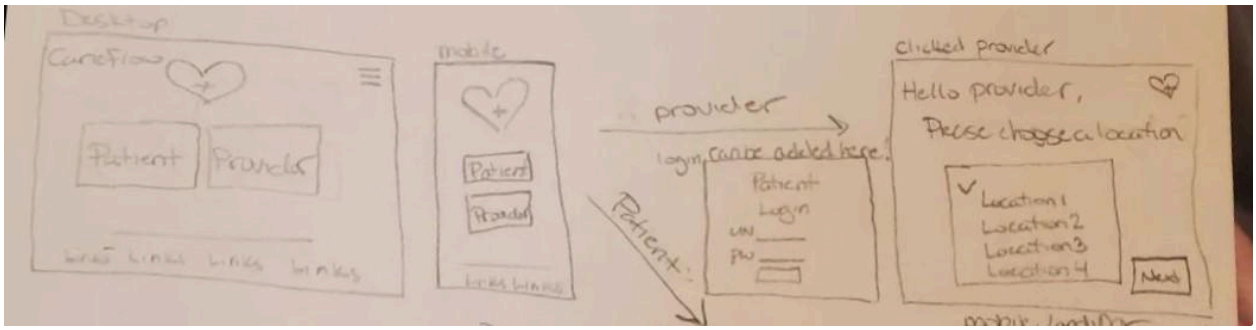
6. **Bill**
    - bill_id (PK)
    - patient_id (FK → Patient)
    - location_id (FK → ClinicLocation) (where service happened)
    - amount, due_date, status (unpaid/paid/void)
    - created_at
    - Relationship: One Patient has many Bills.

7. **PaymentMethod**
    - payment_method_id (PK)
    - patient_id (FK → Patient)
    - type (card/bank)
    - last4, exp_month, exp_year
    - billing_zip
    - active_flag

ERD:

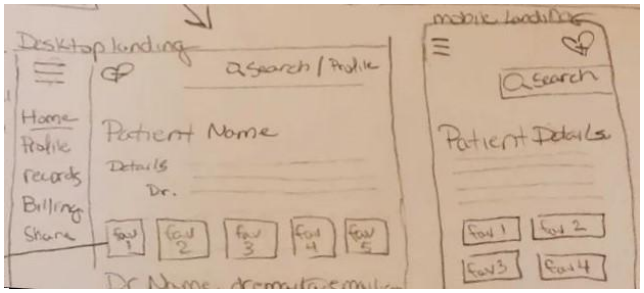
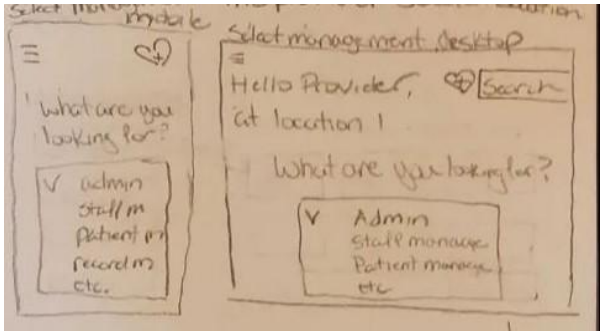
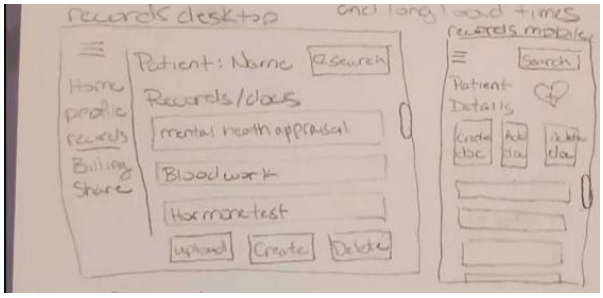
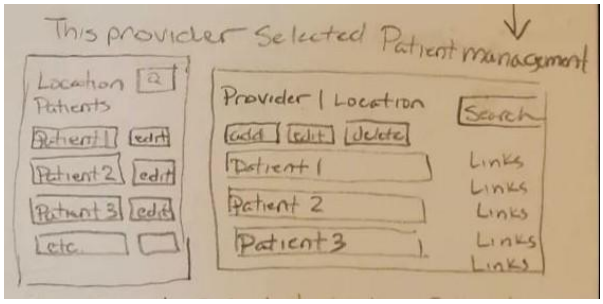
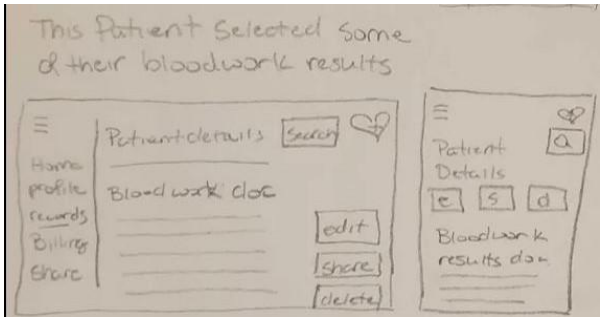
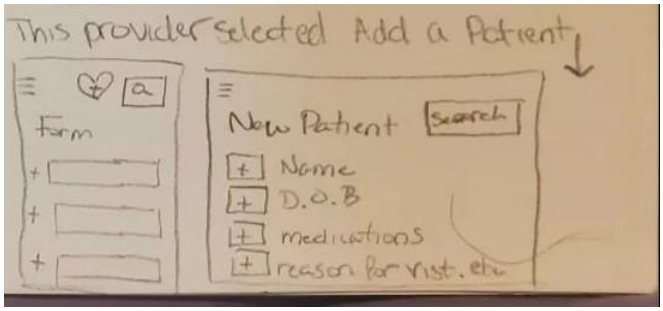


Wireframe/Storyboard:



Above is the initial landing page, which consists of a role selection screen. If the user selects patient, they will be taken down the use case pipeline on the left, and if they select provider they will be taken down the use case pipeline on the right.

Here the user has selected patient, and they are taken to the patients home screen, with all of their generic information, including saved favorites to	Here the user has selected provider, and they are taken to another selection screen where they decide which location they or the patient in question is

cut down on choice overload and fatigue.	located at for treatment. Then they are taken to another selection screen where they choose what use case they need.
	
Next the patient has selected the Records tab, and they are redirected to the records section where they can find all their documents, bloodwork, and test results.	This provider has selected patient management, and is redirected to the patient management menu, where they can see all of their current patients and links to assist in treatment.
	
Next this patient has selected some of their bloodwork results to share with a specific physician, and have editing options and Dr. signoff as well.	This provider has selected Add a patient, which allows them to create a new patient in the system and input their personal information to keep on record.
	

The wireframes included here cover the core functionalities and primary user workflows for both patient and provider roles. For the patient role, the wireframes illustrate dashboard access, profile management, viewing and updating patient records, and document sharing. For the provider role, the wireframes show dashboard access, clinic location selection, patient

management, and adding new patients. Together, these wireframes represent the system's main use cases and demonstrate how each user role interacts with the platform to accomplish their tasks.

Both desktop and mobile versions are included for each wireframe, ensuring that the design is responsive and functional across different screen sizes. Desktop layouts utilize tables, panels, and sidebars for efficient data presentation, while mobile layouts adopt vertical stacks, cards, and hamburger menus to accommodate smaller screens. This approach satisfies the responsive design requirement and illustrates how key actions and workflows remain accessible and usable regardless of device.

Overall, these wireframes provide a sketched representation of the system's interface and workflows while maintaining simplicity appropriate for the project proposal phase. Optional or additional screens, such as provider location selection and adding new patients, further demonstrates workflow depth without complicating the sketches.

System Vision Document:
System Name: CareFlow
Project Type: Healthcare Management Web Application
Course: CSI 4999 - Senior Capstone Project
Term: Winter 2026

Problem Description:
Many healthcare organizations still heavily rely on paper based workflows or multiple disconnected systems to manage clinic locations, staff, patient information, billing, and medical records. These processes make it difficult to keep data up to date and accurate, especially when managing multiple clinic locations. Staff can often spend unnecessary time searching for information, duplicating work, or correcting errors caused by wrong information.

As healthcare organizations continually grow, these issues are becoming more and more significant. There is a need for a centralized system that allows clinics to manage administrative and operational tasks efficiently while still ensuring secure handling of sensitive patient information. A single integrated system will improve accessibility, reduce paperwork and better support coordination across multiple clinic locations.

System Capabilities
Careflow system should be capable of the following:

- Managing multiple clinic locations within a single system
- Assigning and managing staff across different clinic locations
- Creating, updating and maintaining patient information/records
- Storing and managing patient medical information and documents
- Generating and managing patient medical information and documents
- Processing payments and tracking billing status
- Uploading, searching and downloading medical documents

- Enforcing role-based access so users will only see features that are relevant to their role

Business Benefits

It is anticipating that the deployment of CareFlow will provide the following benefits:

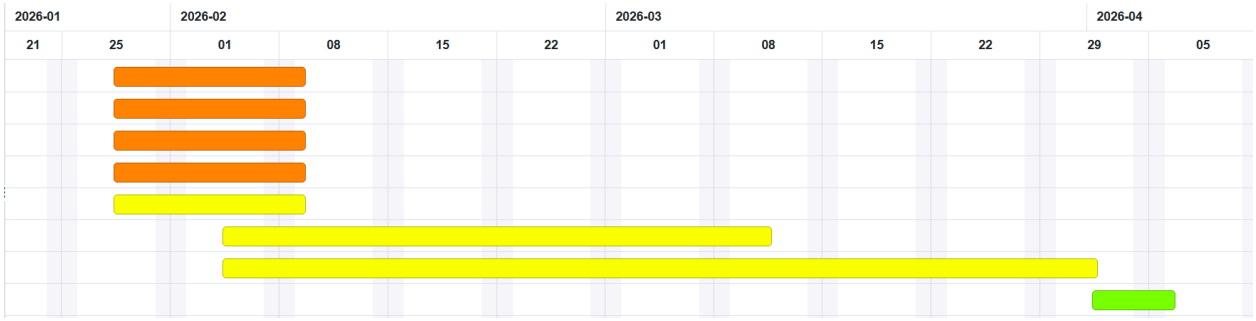
- Reduce reliance on paper based processes and manual record keeping
- Improved data accuracy and consistency across multiple clinic locations
- Increase efficiency for administrative staff by centralizing operations
- Improve accessibility to patient, billing, and document information
- Better coordination between staff, billing and records departments
- Provide an affordable/scalable solution that can grow with healthcare organizations

Work Breakdown Structure and Gantt Chart:

Work Breakdown Structure

1. Discover and understand the details of all aspects of the problem.
   a. Identify and define system functions. ~ 2 hours
   b. Identify and define use cases. ~ 2 hours
   c. Develop workflows and descriptions for the use cases. ~ 3 hours
2. Design the components of the solution to the problem.
   a. Design wireframes (input screens, output screens). ~ 5 hours
   b. Design and build databases (attributes, keys, indexes). ~ 4 hours
   c. Design overall architecture. ~ 4 hours
   d. Design program details. ~ 4 hours
3. Build the components and integrate everything into the solution.
   a. Code and unit test GUI programs. ~ 15 hours
   b. Code and unit test logic programs. ~ 30 hours
4. Perform all system-level tests and then deploy the solution.
   a. Perform final GUI functionality tests. ~ 2 hours
   b. Perform final system functionality tests. ~ 2 hours

	Task Name	ID	Start	End	Duration
	Decide on system fun…	1	2026-01-28	2026-02-09	9 days
	Decide on and define …	2	2026-01-28	2026-02-09	9 days
	Develop workflows	3	2026-01-28	2026-02-09	9 days
	Design wireframes	4	2026-01-28	2026-02-09	9 days
	Build database	5	2026-01-28	2026-02-09	9 days
	Make GUIs	6	2026-02-04	2026-03-11	26 days
	Make back end	7	2026-02-04	2026-04-01	41 days
	Final functionality tests	8	2026-04-01	2026-04-06	4 days



ADDIE Model:

Addie model: analyze, design, develop, implement, evaluate

- **Analyze** other healthcare portals, as well as our own requirements and find a middle ground where the two meet
- **Design** a system that is similar to other successful healthcare programs, but focus on trying to make our design meet the requirements laid out
- **Develop** each moving part and how we can give our features more depth, as well as develop the idea further as a whole
- **Implement** our ideas and features that we designed and developed after analyzing our requirements and other programs
- **Evaluate** how well we met our goals and requirements, user feedback and how we can improve upon our base

Submission

Only one submission for the group is required. Submission includes:

• (70 points) A pdf Project Report (File name example G6.pdf)

• (10 points) A video link of your project idea (3-5 min). Try to focus on the application description and the wireframes first and then you can add whatever you want from the other points. Keep it 3-5 minutes long. One student or more can record the video. This is up to you. You can use YouTube or Loom for the video link. Please do not submit your link in the comment section on Moodle. Include the link in your report.

• Slides of your project idea

• (20 points) Presentation in class: Present your project idea in ,on Feb 5. Each group will present for 10 minutes. All group members should participate. Include the points with * in your slides. You can include other points if needed.